COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S1
DATE PERFORMED:	09/07/2026	**DATE SUBMITTED:**	09/07/2026
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ACTIVITY NO. 1.1 [Basic C++ Programming]

1. PROCEDURE OUTPUT

ILO A

Sections	Answer
Header file declaration section	`#include <iostream>`

Global Declaration Section	`//1`
	`void sum(int fnum, int lnum);`
	
	`//2`
	`void greater(int fvar, int lvar);`
	
	`//3`
	`void logic(bool val1 , bool val2);`

Class Declaration and Method Definition Section	NONE

Main Function	**Problem 1**
	`int main() {`
	`//MAIN CALLER`
	
	`//1`
	`int fnum;`
	`int lnum;`
	
	`//gets the input from the user and assigns it into the given variables`
	`std::cout << "1# welcome to the addition tracker" << std::endl;`
	`std::cout << "enter the first digit" << std::endl;`
	`std::cin >> fnum;`
	`std::cout << "enter the second digit" << std::endl;`
	`std::cin >> lnum;`
	
	`//calls the function that add the numbers up`
	`sum(fnum, lnum);`
	
	Problem 2

```
//2

int fvar;
int lvar;

//gets the input from the user and assigns it into the given variables
std::cout << "2# welcome to the greater or less than checker" << std::endl;
std::cout << "enter first value" << std:: endl;
std::cin >> fvar;
std::cout << "enter the second value" << std:: endl;
std::cin >> lvar;

//calls out the greater function to determine if the first value is lesser or greater than the second value
greater(fvar, lvar);
```

Problem 3

```
//3
bool val1;
bool val2;

//gets the input from the user and assigns it into the given variables
std::cout << "3# welcome to the logic gate displayer" << std::endl;
std:: cout << "type 1 if its true" << std:: endl;
std:: cout << "type 0 if its false" << "\n\n";
std:: cout << "type in the first value: " << std:: endl;
std::cin >> val1;
std:: cout << "type in the second value" << std:: endl;
std::cin >> val2;

//calls out the logic function to determine the result in each logic gate
logic(val1, val2);

  return 0;
}
```

Method Definition

Problem 1

```
//1
void sum(int fnum, int lnum){

//adds the given values and displays the sum
int total = fnum + lnum;
std::cout << "your total: "<< total << "\n\n";


}
```

Problem 2

```
//2
void greater(int fvar, int lvar){

//an if else statement function to check whether its greater than, elss than or equal to
if (fvar > lvar){
    std::cout << fvar << " is greater than " << lvar << "\n\n";
} else if (fvar == lvar){

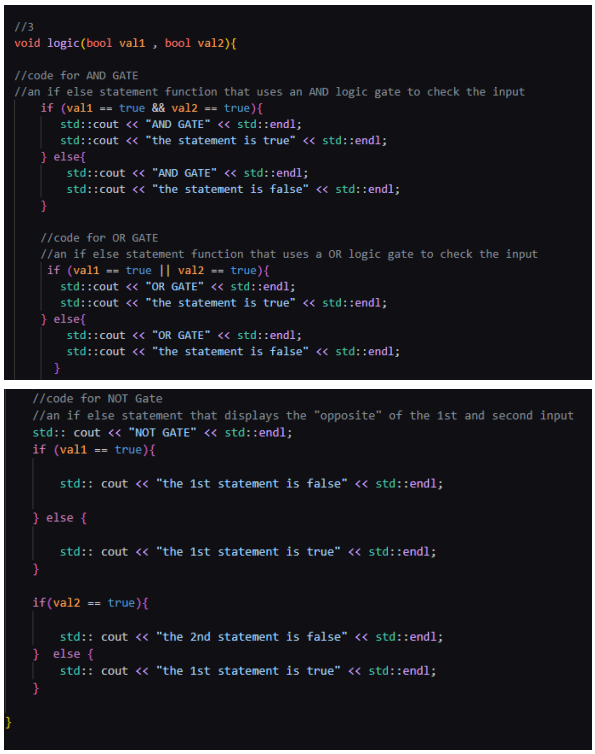
    std::cout<< fvar << " is equal to " << lvar<< "\n\n";

    } else {

    std::cout << fvar << " is less than " << lvar << "\n\n";
}


}
```

Problem 3



```
//3
void logic(bool val1 , bool val2){

//code for AND GATE
//an if else statement function that uses an AND logic gate to check the input
    if (val1 == true && val2 == true){
       std::cout << "AND GATE" << std::endl;
       std::cout << "the statement is true" << std::endl;
    } else{
        std::cout << "AND GATE" << std::endl;
        std::cout << "the statement is false" << std::endl;
    }

    //code for OR GATE
    //an if else statement function that uses a OR logic gate to check the input
     if (val1 == true || val2 == true){
       std::cout << "OR GATE" << std::endl;
       std::cout << "the statement is true" << std::endl;
     } else{
        std::cout << "OR GATE" << std::endl;
        std::cout << "the statement is false" << std::endl;
      }
```

```
   //code for NOT Gate
   //an if else statement that displays the "opposite" of the 1st and second input
   std:: cout << "NOT GATE" << std::endl;
   if (val1 == true){

      std:: cout << "the 1st statement is false" << std::endl;

   } else {

      std:: cout << "the 1st statement is true" << std::endl;
   }

   if(val2 == true){

      std:: cout << "the 2nd statement is false" << std::endl;
   }  else {
      std:: cout << "the 1st statement is true" << std::endl;
   }

}
```

Table 1.1 C++ Code Structure

Problem 1

```
1# welcome to the addition tracker
enter the first digit
30
enter the second digit
40
your total: 70
```

I dont have that much problems with this part apart from being rusty at c++ since I diddn't use this programming language for a while but other than that the program itself is simple I just created a simple addition function and called it in the main caller

Problem 2

```
2# welcome to the greater or less than checker
enter first value
20
enter the second value
40
20 is less than 40
```

```
2# welcome to the greater or less than checker
enter first value
40
enter the second value
40
40 is equal to 40
```

```
2# welcome to the greater or less than checker
enter first value
80
enter the second value
45
80 is greater than 45
```

For this problem, again not that much of a struggle apart from forgetting the simple c++ parts and that includes the if-else statement which was used in this function since the output can either be greater , less or equal to the value.

Problem 3

```
3# welcome to the logic gate displayer
type 1 if its true
type 0 if its false

type in the first value:
1
type in the second value
0
AND GATE
the statement is false
OR GATE
the statement is true
NOT GATE
the 1st statement is false
the 1st statement is true
```

```
3# welcome to the logic gate displayer
type 1 if its true
type 0 if its false

type in the first value:
1
type in the second value
1
AND GATE
the statement is true
OR GATE
the statement is true
NOT GATE
the 1st statement is false
the 2nd statement is false
```

```
3# welcome to the logic gate displayer
type 1 if its true
type 0 if its false

type in the first value:
0
type in the second value
0
AND GATE
the statement is false
OR GATE
the statement is false
NOT GATE
the 1st statement is true
the 1st statement is true
```

For this problem here this is where I had to use my logical thinking on how to output all the gates, on paper it looks difficult but once I knew how to output them its a moment of making the functions. I do have to admit this one is time consuming because I had to not only relearn the if-else statement but how to use them properly, theres also the logic gates which I completely forgot about and I'll be honest it made this whole function easy which I used in the program.

Table 1.2 C++ Output and Reflection

ILO B

Sections	**Answer**
Header file declaration section	`#include <iostream>`
Global Declaration Section	NONE

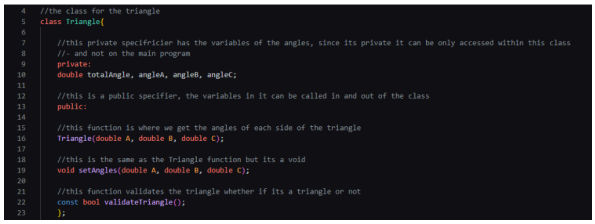
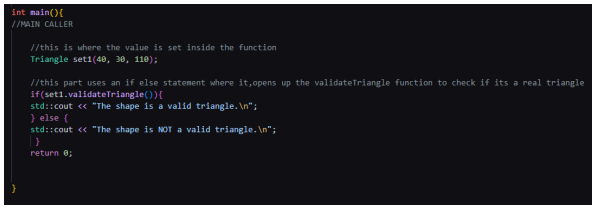
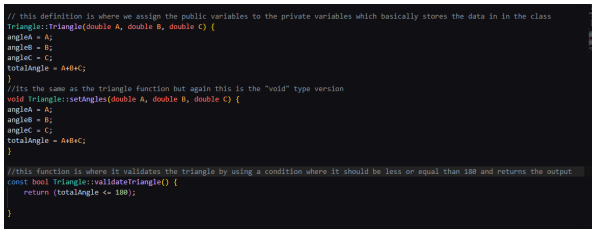
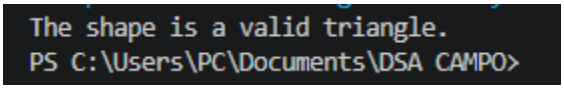
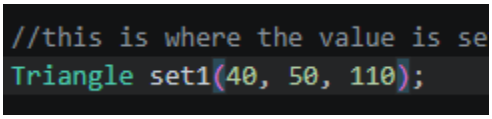
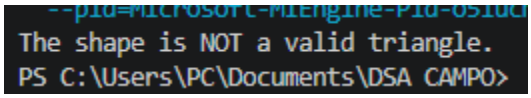
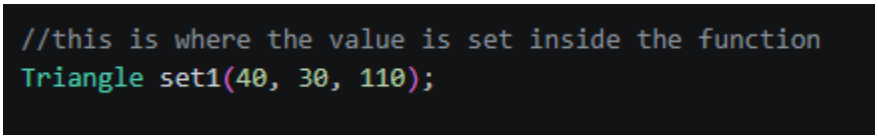
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

Table 2.1 C++ Code Structure









For this activity it was already posted on the laboratory manual, including the codes meaning all I had to do was just copy it. But I did make sure to understand it. From what i've learned a class is where multiple functions are stored in, you can also make sure the variables there can be accessed by the main caller

with the use of the “public” and “private” specifiers. I did have a hard time learning this because I wasn’t familiar with classes and I got confused on the Triangle() function and what differs it from the void setAngles() function. Little did I knew they perform the same function but one is a void and another one is the main constructor

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

(Problem 1 - 3 is created in one program)

1. Create a C++ program to swap the two numbers in different variables.

Code Screenshot:

```
#include <iostream>

//1

//the global declaration section to swap 2 values
void swap(int num1, int num2);
```

```
int main(){
//MAIN CALLER

    //1
    //it assigns the value to the variable inside the function
    int num1 = 7;
    int num2 = 14;

    //calls out the function
    swap(num1 , num2);
```

Method Definition:

```
//1

//this function uses another variable to store the data of the first value so that the 2nd value can be the new first value
//it also displays the before and after the switch
void swap(int num1, int num2){
    int extra;
    std:: cout << "before: " << num1 << ", " << num2 << std:: endl;

    extra = num1;
    num1 = num2;



   std:: cout << "after: " << num1 << ", " << extra << std:: endl;
}
```

OUTPUT:

```
before: 7, 14
after: 14, 7
```

This program itself is simple but I for me I had trouble figuring out how to swap the value of the variables but in the end I just created a new variable to store the first value in and displayed it here. Other than that there are no issues like I said this program itself is simple once I figured it out its smooth sailing from there

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Code Screenshot:

```
//2

//Global declaration section
void convert(float far, float deg);
```

Int Main():

```
  //2
  //it assigns the value to the variable inside the function
  float far;
  float kel;
  //calls out the function
  convert(far, kel);
```

Method Definition:

```
//2 this function asks the user for the temperature in kelvin which is trasnfered into a convertion formula that is assigned-
//to the farenheit variable and then it outputs the temperature in farenheit
void convert(float far, float kel){



    std:: cout << "welcome to kelvin to farenheit converter" << std:: endl;
    std:: cout << "enter the temperature in kelvin: ";
    std:: cin >> kel;

    far = (((kel - 273.15)* 1.8) + 32) ;

    std:: cout << "the temperature in farenheit is: " << far<< "\n\n";

}
```

Output:

```
welcome to kelvin to farenheit converter
enter the temperature in kelvin: 200
the temperature in farenheit is: -99.67
```

This code was similar to the program I did before but instead of kelvin we used celsius, right now I forgot the process I did before but after doing the previous activities in this HOA I recalled how I did it which is just to assign the fahrenheit variable with the conversion formula. Typing in the formula itself diddn't cause me any troubles surprisingly (probably because of my experiences in MATLAB) but overall this program wasn't hard to make

3. Create a C++ program that has a function that will calculate the distance between two points.

Code Screenshot:

```
//3

//Global declaration section
void distance(float point1, float point2 );
```

Int Main():

```
//3
//it assigns the value to the variable inside the function
float point1;
float point2;
//calls out the function
distance(point1, point2);
```

Method Definition:

```
//3 this function asks the user for the first point and second point, it then subtracts the 2nd point to the 1st point to get the-
//distance (which is assigned to fpoint) and it is then outputed
void distance(float point1, float point2 ){

    float fpoint;
    std:: cout << "welcome to the distance checker" << std:: endl;
    std:: cout<< "enter the first point (in meters): " ;
    std:: cin >> point1;
    std:: cout << "enter the second point (in meters): ";
    std:: cin >> point2;
    fpoint = point2 - point1;
    std:: cout << "the distance between point 1 and point 2 is: " << fpoint;
}
```

Output:

```
welcome to the distance checker
enter the first point (in meters): 80
enter the second point (in meters): 100
the distance between point 1 and point 2 is: 20
```

This program is simple and at this point in the activity I had no trouble programing this I just took the 2 points from the user subtracted the bigger value with lower value to get the distance of the 2 points, its a simple input-output program

4. Modify the code given in ILO B and add the following functions:

Code Screenshot:

```cpp
#include <iostream>

//Class Declaration and Method Definition Section

//the class for the triangle
class Triangle{

    //this private specifier has the variables of the angles, since its private it can be only accessed within this class
    //- and not on the main program
    private:
    double peri, area, angleA, angleB, angleC, base, height, sideA, sideB, sideC;

    //this is a public specifier, the variables in it can be called in and out of the class
    public:
    Triangle(double A, double B, double C, double b, double h, double sA, double sB, double sC);


  //a) the function to identify the type of triangle
  void TypeOfTriangle();

  //b) the function to identify the area of the triangle
  void AreaOfTriangle();

  //c) the function to identify the perimeter of the triangle
  void PerimeterOfTriangle();
   };

```

Int Main():

```
int main(){
//MAIN CALLER

 //variables
double A;
double B;
double C;
double b, h, sA, sB, sC;

//this section asks the user for the needed inputs (height, base, side and angles)
    std:: cout << "welcome to the triangle perimeter, area and type of angle checker \n";
    std:: cout << "enter the 1st angle \n";
    std:: cin >> A;
    std:: cout << "enter the 2nd angle \n";
    std:: cin >> B;
    std:: cout << "enter the 3rd angle \n";
    std:: cin >> C;
    std:: cout << "enter base: ";
    std:: cin >> b;
    std:: cout << "enter height: ";
    std:: cin >> h;
    std:: cout << "enter side A: ";
    std:: cin >> sA;
    std:: cout << "enter side B: ";
    std:: cin >> sB;
    std:: cout << "enter side C: ";
    std:: cin >> sC;
```

```
   //this is where it calls the constructor and assigns the value of the variables in the main caller to the triangle class
   Triangle set1(A, B, C, b, h, sA, sB, sC);

   //outputs the triangle components by calling the functions
   std:: cout << " \n";
   std:: cout << "TRIANGLE COMPONENTS \n";
   set1.TypeOfTriangle();
   set1.AreaOfTriangle();
   set1.PerimeterOfTriangle();
   return 0;


}
```

Method Definition:

```
//function/class definitions

//defines the triangle class and the functions inside it
//assigns the private variables with the public variables, it also assigns the formula to the perimeter and area
Triangle::Triangle(double A, double B, double C, double b, double h, double sA, double sB, double sC) {

angleA = A;
angleB = B;
angleC = C;
base = b;
height = h;
sideA = sA;
sideB = sB;
sideC = sC;

peri = sA+sB+sC;
area = ((h * b)/ 2);
}
```

```
//this function gets the area from the triangle class and outputs it
void Triangle:: AreaOfTriangle(){

    std:: cout << "the area of this triangle: " << area << std:: endl;

}
//this function gets the perimeter from the triangle class and outputs it
void Triangle:: PerimeterOfTriangle(){
    std:: cout << "The perimeter of this triangle: " << peri << std:: endl;

}
//this function gets the type of the triangle class and outputs it using an if else statement to determine what type it is
 void Triangle::TypeOfTriangle(){
    if( angleA < 90 && angleB < 90 && angleC < 90){

        std:: cout << "this triangle is an acute triangle \n";
    } else if (angleA > 90 || angleB > 90 || angleC > 90 ){

        std::cout<< "this triangle is a obtuse Triangle \n";
    } else{
        std:: cout << "other triangles \n";
    }
 }
```

Output:

```
welcome to the triangle perimeter, area and type of angle checker
enter the 1st angle
30
enter the 2nd angle
60
enter the 3rd angle
30
enter base: 50
enter height: 60
enter side A: 40
enter side B: 40
enter side C: 40

TRIANGLE COMPONENTS
this triangle is an acute triangle
the area of this triangle: 1500
The perimeter of this triangle: 120
```

For me this code was difficult especially part C where I have to determine what type of angle it is. It would be easy if it wasnt for one thing which is the components of an obtuse angle where it needs atleast one angle for it be obtuse and that for me I just cant implement it into this program, first I used nested list but it was too complicated for me to do so I think of other ways and eventually I realized that a simple OR condition can solve my problem so I used that for the TypeOfTriangle() function. As for the other functions it wasn't that much of an issue, setting them up tho gave me another challenge. Since Im not familiar with class I had to learn how it works and it made me realize we really do need the significance of the "private" and "public" but eventually I pulled through and finished this program with output I wanted to have

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To get examples and to explain each line of code**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I had to use it on supplementary activity: problem 4 and in ILOB**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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